Junior Makers · Class 13

Step Counter
Fitness Tracker

Build a gadget that counts every step you take!

Accelerometer **Reading graphs** **Threshold rules**



Did you know?
Doctors say kids should be active for at least 60 minutes every day. Step counters help people check.

Let's think...

How does a watch know you took a step?

7,482 steps today!

Nobody follows you around counting. The watch has no camera on your feet. Yet it knows when you walk, run or sit still. What is inside that **feels** you move?



A tiny camera?

A spring?

A motion sensor!

Today: we build our own step counter with a **motion sensor**, show the steps on a screen, and test how accurate it is.

Learning objectives

By the end of this class I can...

1	**Feel motion:** explain how an **accelerometer** feels movement in 3 directions	**Proof:** say which axis changes when I move the sensor up and down
2	**Read data:** find the **steps** in a graph of walking data	**Proof:** point to the peaks in my Serial Plotter graph
3	**Count:** count a step with a **threshold** rule that counts each step only once	**Proof:** my OLED counts up when I walk and not when I stand
4	**Test:** test my tracker against **50 real steps** and work out the error	**Proof:** my accuracy table is filled in

Objective 1 · Feel motion · video

Video time!



How a Smartphone Knows Up from Down

engineerguy (Bill Hammack) · about 4 minutes · youtu.be/KZVgKu6v808

Watch for...

1. What tiny thing **moves** inside the chip when the phone moves?
2. Why does the sensor still feel something when the phone is **still**?
3. Name **one job** the accelerometer does in a phone.

Scan to watch

After the video: Hold up a phone or a book. Show your partner which way is X, Y and Z, and explain one thing the sensor feels.

Objective 1 · Feel motion

An accelerometer feels pushes

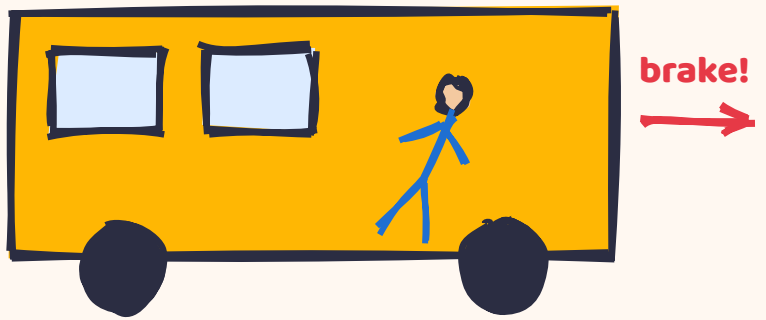
The idea

Inside the chip is a **tiny weight on springs**, thinner than a hair. When you speed up, slow down or bounce, the weight **lags behind** and squeezes the springs. The chip measures that squeeze.

It measures in **3 directions**: X (side to side), Y (forward and back), Z (up and down).

Surprise
Even when still, the sensor feels about 9.8 pushing up: that is gravity!

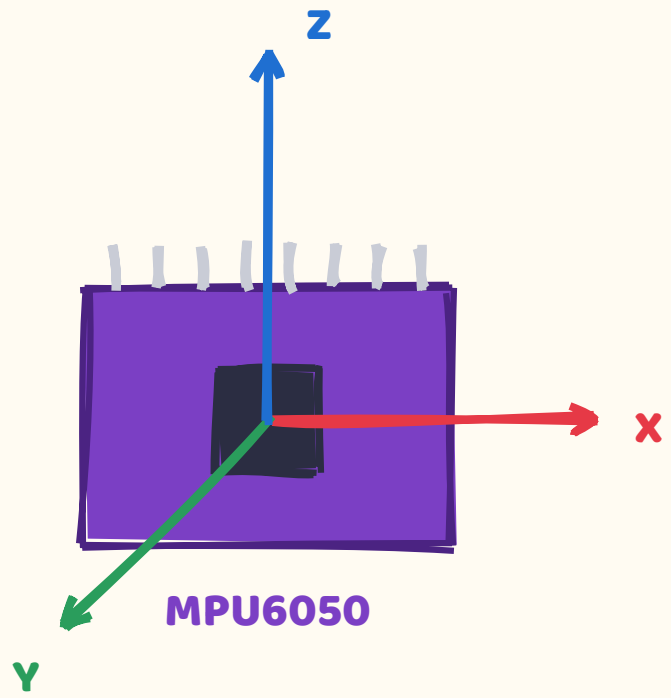
Example: standing on a bus



When the bus **brakes**, your body **leans forward**. When it **speeds up**, you lean back. You are like the tiny weight: you feel the push even with your eyes closed.

Objective 1 · Feel motion · meet the sensor

The MPU6050 motion sensor



Do this	What changes
Lie it flat, still	X ≈ 0 · Y ≈ 0 · **Z ≈ 9.8**
Tilt it left	X changes
Push it forward	Y changes
Bounce it up and down	**Z** jumps up and down

Also a gyro!

The MPU6050 also measures **turning** (gyroscope). Today we only use the accelerometer.

Why 9.8?

Gravity pulls everything down with 9.8 m/s². The sensor feels the table pushing back up. (Some blocks show 1.0 g instead: same thing!)

Real world

Accelerometers are everywhere



Smartwatches
Count steps and notice if you fall



Phones
Turn the screen when you rotate the phone



Car airbags
Feel a crash and open in a split second



Game controllers
Turn your swing into a tennis shot

? Name a time an accelerometer might have helped you today.

Tiny!
The sensor in a watch is smaller than a grain of rice and costs less than a sweet.

Objective 2 · Read data

Find the steps in the data

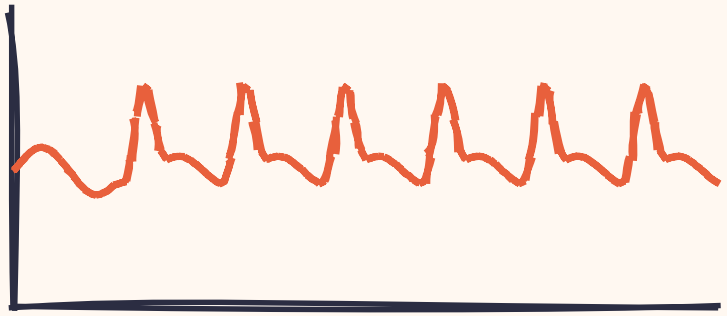
The idea

The Arduino reads the sensor about **20 times a second**. We draw the readings as a **graph** with the **Serial Plotter**.

Standing still: a flat wobbly line near 9.8.
Walking: the line goes up and down. **Each peak is one step!**

One number
We add X, Y and Z together into one 'shake' number, so it works however you hold the tracker.

Example: counting waves at the beach



standing **walking: each peak = 1 step**

Watch the sea and count every big wave that reaches the sand. You ignore the tiny ripples. Our step counter counts the **big peaks** and ignores small wobbles.

Objective 3 · Count

A threshold decides YES or NO

The idea

A **threshold** is a line we choose. Above the line = **yes, a step**. Below = **no**.

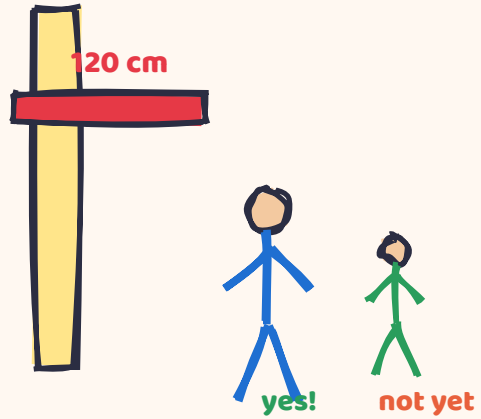
Look at your graph: standing stays near **9.8**, walking peaks reach about **12 to 14**. So a good threshold is about **11.5**.

Too low: counts wobbles. Too high: misses gentle steps.

Yours is unique

Every person walks differently. Your threshold might be 11 or 12.5. Test and choose!

Example: a ride height bar



At a theme park, a sign says **'You must be this tall'**. 120 cm is the threshold. Taller = yes, you can ride. Shorter = not yet. The bar does not care by how much!

Objective 3 · Count

Count each step only ONCE

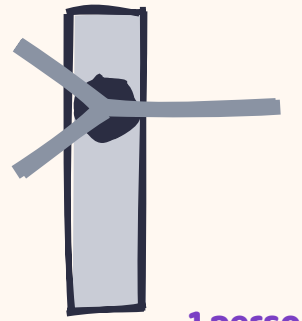
The idea

One peak stays above the threshold for several readings. If we count every reading, one step becomes **4 steps**!

Fix it with a **flag** called **stepping**:

- shake goes **above 11.5** and not stepping → add 1 step, stepping = true
- shake drops **below 10.5** → stepping = false (ready for the next step)

Example: a turnstile



1 person = 1 click

A turnstile at a stadium clicks **once** per person. It will not click again until the arm has turned all the way round. Our flag is the arm: it must reset before the next count.

Objective 4 · Test

How accurate is my tracker?

The idea

Engineers **test** against something they trust. Walk exactly **50 steps** while a partner counts out loud.

Error = tracker - real
Error % = error ÷ real × 100

Example: tracker 46, real 50 → error -4 → **8% too low**.

Target
Under 5% error is great for a homemade tracker!

Example: a bathroom scale



compare to check

If a scale says you weigh 2 kg more than the doctor's scale, it has an **error**. You only know because you **compared** it with a better one. Our 'better scale' is a partner counting carefully.

Quick quiz

True or false?

Statement	Circle
1. An accelerometer measures temperature.	T / F
2. When the sensor is still, it reads 0 on every axis.	T / F
3. Each peak in the walking graph is about one step.	T / F
4. A threshold that is too low counts extra steps.	T / F
5. The OLED and MPU6050 can share the SDA and SCL wires.	T / F

Answers

Cover this! 1 F · 2 F ($z \approx 9.8$, gravity) · 3 T · 4 T · 5 T



Part 2 · Make it

Now let's put it to work!

You know the ideas. Time to play, build and test. Use these slides while you work:

1. The water-cup walk
2. Our fitness tracker kit (per group)
3. Wire the tracker
4. Be a data detective
5. The step counter code
6. The 50-step test
7. Fitness tracker challenges

Class game

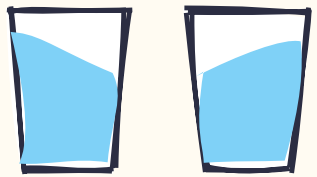
The water-cup walk

How to play

1. Each team gets a cup half full of water.
2. Walk slowly across the room holding the cup out in front.
3. Watch the water: it **sloshes once** with every step.
4. A partner counts the sloshes. Another counts the real steps.
5. Do the numbers match? Try walking faster!

Then discuss

Why does the water move with every step? What would a step counter 'feel' if you were in a car instead?



Big idea

Every step gives your body a little up-and-down bump. The sensor feels each bump, just like the water.

Build it

Our fitness tracker kit (per group)



Arduino Uno
The brain + USB cable



MPU6050
Motion sensor



OLED screen
Shows the steps



Push button
Reset to 0 (+ 10k)



LED + 220 Ω
Blinks each step



Buzzer
Beeps at your goal

Also: breadboard, jumper wires, laptop, a power bank and a pencil case or box to carry the tracker while walking.

No hardware?
Build and test it first in the free Wokwi simulator: it has an MPU6050 you can shake with sliders!

MAKE

Objective 1 · Feel motion · Build step 1 of 3

Wire the tracker

Part	Connect to
MPU6050 VCC / GND	5V / GND
MPU6050 SDA / SCL	A4 / A5
OLED VCC / GND	5V / GND
OLED SDA / SCL	A4 / A5 (same pins!)
Button	pin 2 and 5V · 10k from pin 2 to GND
LED (+220 Ω) / Buzzer	pin 12 / pin 11

Two parts, same 2 wires?

Yes! **SDA** and **SCL** make a **shared bus** called **I2C**. Each part has its own **address**, like houses on one street: MPU6050 is **0x68**, OLED is **0x3C**.

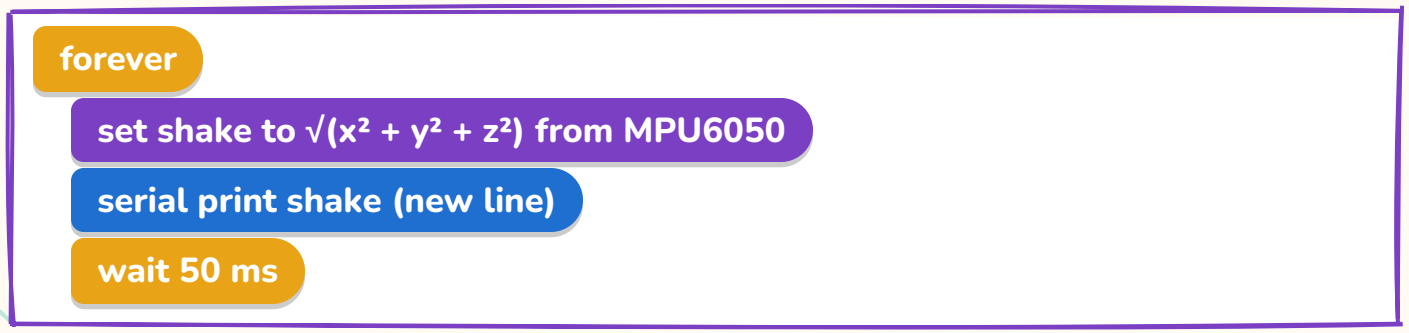
Check

Run an I2C scanner. You should see two addresses. Only one? Check that part's wires.

Objective 2 · Read data · Build step 2 of 3

Be a data detective

Do this	Predict the graph	What really happened
Stand very still	flat, about 9.8	
Walk 10 steps		
Jump 3 times		
Shake it hard		



Open the plotter

Upload, then **Tools → Serial Plotter** (or the graph button in ArduinoBlocks). Carry the laptop and tracker together!

Find the number

Write down how high your walking peaks go. You will need it for the threshold!

Objective 3 · Count · Build step 3 of 3

The step counter code

```
when Arduino starts
    set steps to 0 · set stepping to false · OLED start
forever
    set shake to √(x² + y² + z²)
    if shake > 11.5 and stepping = false
        change steps by 1 · set stepping to true · LED blink
    if shake < 10.5 → set stepping to false
    if button pressed → set steps to 0
    if steps = 100 → beep (goal!)
    OLED show "Steps:" and steps · wait 50 ms
```

Read it as a story

Feel the shake. Big shake and not already stepping? That is a new step! Shake calmed down? Get ready for the next one.



The OLED shows the total. The **button** resets it. The **LED** blinks with every step.

Use variables

steps, stepping and shake are boxes that remember numbers.

Objective 4 · Test · test it

The 50-step test

Test	Real steps	Tracker says	Error	Error %
Slow walk	50			
Normal walk	50			
Fast walk	50			
Up the stairs	20			

Compare with a friend: swap trackers and walk the same 50 steps. Whose tracker is more accurate? Why might that be?

Make it fair

Same person, same route, same way of holding the tracker every time.

Fix it

Counting too many? Raise the threshold. Too few? Lower it. Test again!

Level up!

Fitness tracker challenges

Choose your challenge



Daily goal: show a star on the OLED when you reach your goal (try 200 steps).



Distance: multiply steps by your stride length (about 0.5 m) to show metres walked.



Lazy alarm: beep if no step happens for 1 minute: time to move!



Run detector: show WALK or RUN depending on how high the peaks are.

Think like an engineer

Why might a step counter count steps when you are riding in a bumpy car?

Boss level

Measure your own stride: walk 10 steps, measure the distance, divide by 10.

Objective check

Can I really do it? Prove it.

01	Move your sensor up and down. Which axis changes? Why does Z show 9.8 when still?	○ ○ ○
02	Point to 3 steps in your Serial Plotter graph.	○ ○ ○
03	Explain why we need the **stepping** flag. What would go wrong without it?	○ ○ ○
04	Read out your 50-step test result and your error %.	○ ○ ○

Rate yourself

Colour one circle each. Red: not yet. Amber: with help. Green: I can explain it to a friend.

Teacher: Pick 2 groups with very different thresholds and ask them to explain why.

Show what you know

What did we learn?

- [x] An accelerometer feels pushes in X, Y and Z
- [x] Still, it feels gravity: about 9.8
- [x] Each peak in walking data is one step
- [x] A threshold plus a flag counts each step once
- [x] Testing against real steps shows our error

At home

Count your steps from home to school one day. Then guess how many steps you walk in a week.



Exit ticket: why does our tracker need the stepping flag?

Next class: build a Weather Report Webpage!